AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS 2021

Computer Science

Total Time: 2 hours 10 minutes

Total Marks: 65 (50-Theory & 15-Alternate to Practical)

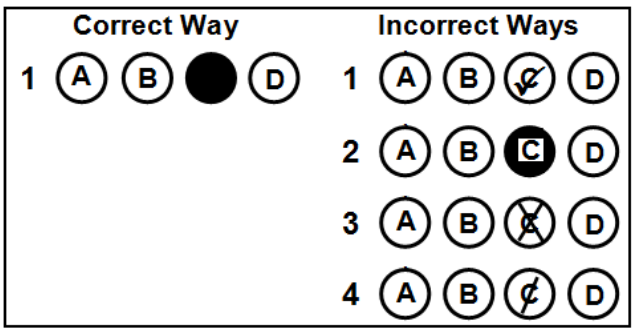
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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 65 only.
4. Question Distribution:

Theory	Alternate to Practical (ATP)
50 MCQs	15 MCQs

5. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

6. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
7. DO NOT write anything in the answer grid. The computer only records what is in the circles.
8. The marks obtained on the 50 MCQs will be equated to the total marks of 85 for the theory examination results.
9. You may use a scientific calculator if you wish.

THEORY (Questions 1-50)

1. A C++ program usually starts with the

   A. cin keyword.
   B. cout keyword.
   C. main function.
   D. preprocessor directive.

2. The single line comment in a C++ program starts with

   A. \\
   B. //
   C. /*
   D. *

3. Areeba has written a program in C++ that

   - defines a string constant named as FullName which stores the name Akram Ali
   - displays the name using string constant

   Which of the following programs she must have written?

A	B
C	D

A
```
#include <iostream>
using namespace std;
#define FullName "Akram Ali"
int main()
{
  cout << "FullName";
  return 0;
}
```

B
```
#include <iostream>
using namespace std;
#define FullName "Akram Ali"
int main()
{
  cout << FullName;
  return 0;
}
```

C
```
#include <iostream>
using namespace std;
#include <FullName Akram Ali>
int main()
{
  cout << FullName;
  return 0;
}
```

D
```
#include <iostream>
using namespace std;
#define FullName Akram Ali
int main()
{
  cout << "FullName";
  return 0;
}
```

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4. Consider the given program.

```
#include <iostream>
using namespace std;
int main()
{
int J = 17;
if (J == 18)
cout << "The value of J is 17";
cout << "The value of J is 18";
return 0;
}
```

The output of this program is

The value of J is 18	The value of J is 17
A	B
Error in program	Program executes but no output is displayed
C	D

5. Which of the following MUST be present in every C++ program?

A. Loop
B. Array
C. Variable
D. Function

6. The memory address of a variable is stored in a/ an

A. object.
B. pointer.
C. function.
D. constructor.

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7. Consider the given program.

```
 #include <iostream>
using namespace std;
int main()
{
int K = 2;
if (K == 1)
{
cout <<"K is 1";
}
else if (K == 2)
{
cout << "K is 2";
}
else
{
cout <<"value of K is unknown";
}
  return 0;
}
```

The code that replaces the shaded code in the given program and still gives the same output is

switch (K)
{
case 1:
cout <<"K is 1";
break;
case 2:
cout <<"K is 2";
default:
cout <<"value of K is unknown";
}
	switch (K)
	{
	case 1:
	cout <<"K is 1";
	case 2:
	cout <<"K is 2";
	break;
	default:
	cout <<"value of K is unknown";
	}

A	B

switch (K)
{
case 1:
cout <<"K is 2";
break;
case 2:
cout <<"K is 1";
break;
default:
cout <<"value of K is unknown";
}
	switch (K)
	{
	case '1':
	cout <<"K is 1";
	break;
	case '2':
	cout <<"K is 2";
	break;
	default:
	cout <<"value of K is unknown";
	}

C	D

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8. Consider the given statement.

   float P = 0.0000643091;

   The value of P will be displayed in the output as

   A. 0.000064
   B. 6.43091e-005
   C. 6.43091e+005
   D. 0.0000643091

9. The escape sequence that can be used as an alternative to **endl** is

   A. \a
   B. \b
   C. \t
   D. \n

10. An example of manipulators is

    A. setw.
    B. cout.
    C. sizeof.
    D. strcmp.

11. Which of the following is NOT a reserved word of the C++ programming language?

    A. as
    B. if
    C. or
    D. do

12. Following variables are declared in a program.

```
short int a;
long int b, c;
double d;
```

    The total memory storage (in bytes) occupied by these variables is

    A. 14
    B. 18
    C. 26
    D. 28

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Use the given program to answer Q.13, Q.14, Q.15 and Q.16.

```
#include <iostream>
using namespace std;
int main()
{
int j;
for(j=5; j>1; j--)
cout << j << " ";
cout << endl;
return 0;
}
```

13. The manipulator used in this program is

A. endl.
B. return.
C. iostream.
D. namespace.

14. The code line that should replace **cout << j << " ";** in the program to get the given output is

```
27 18 11 6
```

A. cout << j * (j + 1) - 3<< " ";
B. cout << (j * 7) - 8<< " ";
C. cout << j + 4 * 3<< " ";
D. cout << j * j + 2<< " ";

15. Consider the given code from the program.

```
for(j=5; j>1; j--)
cout << j << " ";
```

Which of the following codes should replace it so that the loop should repeat the code 12 times?

for(j=1; j<25; j++)
{
cout << j << " ";
j++;
}
	for(j=1; j<=25; j++)
	cout << j << " ";
	j++;

A	B

for(j=1; j<=25; j++)
{
cout << j << " ";
j = j + 1;
}
	for(j=1; j=25; j++)
	cout << j << " ";
	j = j + 1;

C	D

16. Consider the given code from the program.

```
int j;
for(j=5; j>1; j--)
cout << j << " ";
```

If this code is rewritten by using **while loop**, then the code that should replace it is

int j = 1;
while (j < 5)
{
cout << j << " ";
j = j - 1;
}
	int j = 1;
	while (j > 5)
	{
	cout << j << " ";
	j = j + 1;
	}

A	B

int j = 5;
while (j > 1)
{
cout << j << " ";
j = j - 1;
}
	int j = 5;
	while (j < 1)
	{
	cout << j << " ";
	j = j - 1;
	}

C	D

17. The total memory storage occupied by five variables (i, j, k, m and n) in a program is 23 bytes.

Based on this memory storage, the variables must have been declared as

char i, j; double k; long int m; short int n;	char i; double j, k; long int m; short int n;
A	B
char i; double j; long int k, m; short int n;	char i; double j; long int k; short int m, n;
C	D

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Use the given program to answer Q.18 and Q.19.

```
#include <iostream>
using namespace std;
int main()
{
int @num1 = 10;
cout << @num1;
return 0;
}
```

18. The complier shows an error in the variable name used in this program.

    The compiler will execute the variable name if it is written as

    A. 1num
    B. num@1
    C. _num1
    D. num#1

19. Which of the following options is CORRECT about the argument and return value of the main function?

	Contains Argument	Returns Value
A	Yes	Yes
B	Yes	No
C	No	Yes
D	No	No

20. The expression that contains two relational operators is

    A. (x + y) <= (y + z)
    B. (x == y) && (y != z)
    C. (x && y) || (y && z)
    D. (++x / --y) >= (++y / --z)

21. The expression that uses the maximum number of logical operators is

    A. (x * y) <= (y % z)
    B. (x || y) && (y != z)
    C. (x <= y) || !(y && z)
    D. (++x >= --y) == (++y || --z)

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22. Consider the given expression.

(x && y) || (y && z)

The expression evaluates TRUE (1) when the values of x, y and z are

A. x = 0, y = 1 and z = 0.
B. x = 1, y = 0 and z = 0.
C. x = 0, y = 1 and z = 1.
D. x = 1, y = 0 and z = 1.

23. Consider the given expression.

(!x || y) && (y || !z)

The expression evaluates FALSE (0) when the values of x, y and z are

A. x = 1, y = 1 and z = 1.
B. x = 0, y = 0 and z = 0.
C. x = 0, y = 1 and z = 0.
D. x = 1, y = 0 and z = 1.

24. Consider the given values of x, y and z.

x = 0, y = 1, z = 1

The expression that evaluates TRUE (1) for these values of x, y and z is

A. (!x || y) && (x || z).
B. (x || !y) && (x || z).
C. (x && !y) || (x && !z).
D. (x && !y) || (!x && !z).

25. An array is declared by specifying its

A. data type followed by its size and then its name.
B. data type followed by its name and then its size.
C. name followed by its data type and then its size.
D. name followed by its size and then its data type.

26. The size of an array is specified inside

A. round brackets.
B. square brackets.
C. curly brackets.
D. angle brackets.

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27. When values from an array are read by using a loop, then which of the following things, related to the array element, changes or remains constant with each iteration of the loop?

	Index Number of Array Element	Name of Array Element	Value of Array Element
A	Remains Constant	Changes	Remains Constant
B	Remains Constant	Remains Constant	Changes
C	Changes	Remains Constant	Remains Constant
D	Changes	Remains Constant	Changes

28. Which of the following statements is TRUE about a string variable?

A. It is a one-dimensional array of the integer data type.
B. It is a two-dimensional array of the integer data type.
C. It is a two-dimensional array of the character data type.
D. It is a one-dimensional array of the character data type.

29. Which of the following is automatically appended at the end of each string?

A. '0'
B. '\0'
C. '/0'
D. '\0/'

30. The following statement shows the initialisation of a string.

```
char stu_name[7] = "Fatima";
```

Another way to initialise this string is

A. char stu_name[7] = ['F' 'a' 't' 'i' 'm' 'a'];
B. char stu_name[7] = {'F' 'a' 't' 'i' 'm' 'a'};
C. char stu_name[7] = ['F', 'a', 't', 'i', 'm', 'a'];
D. char stu_name[7] = {'F', 'a', 't', 'i', 'm', 'a'};

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Use the given program to answer Q.31, Q.32 and Q.33.

```
#include <iostream>
using namespace std;
void G(int x, int y, int z)
{
   x = x + y;
   y = z + x;
   z = x + y;
   cout << x << endl;
   cout << z - y << endl;
   cout << x + z;
}
int main()
{
   int a = 2, b = 3, c = 4;
   G(a,b,c);
   return 0;
}
```

31. A component of the user-defined function that is NOT present in this program is

    A. function header.
    B. calling the function.
    C. function declaration.
    D. invoking the function.

32. The number of actual parameters in this program is

    A. two.
    B. three.
    C. five.
    D. six.

33. The output of the given program is

5 5 19	2 1 6
A	B
5 9 14	2 3 4
C	D

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34. Sana has written a C++ program. To perform a specific task, she has written a piece of code in the program in 5 different places.

A programming concept that she could have used to write the code once and use it in multiple places is

A. loop.
B. array.
C. pointer.
D. function.

35. Consider the given list of functions.

I. div()
II. add()
III. sub()
IV. abs()

The built-in functions in this list are

A. I and II.
B. I and IV.
C. II and III.
D. III and IV.

36. A return type that a function may have is

A. loop.
B. array.
C. object.
D. switch-case.

37. Other than function prototype, how many components of a function is/ are there?

A. One
B. Two
C. Three
D. Four

38. A function prototype ends with a

A. colon.
B. semicolon.
C. parenthesis.
D. number sign.

39. For a user-defined function, a function prototype is required in a C++ program when the function is

A. defined after the main function body.
B. invoked after the main function body.
C. defined before the function prototype.
D. invoked before the function prototype.

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40. The CORRECT way to declare a pointer variable is

A. int x*;
B. int *x;
C. int &x;
D. int x&;

41. A statement that stores the address of variable G into pointer H is

A. G = H&;
B. G = &H;
C. H = &G;
D. H = G&;

42. Consider the given statement.

cout << *F;

The output of this statement will be

A. value stored in pointer variable F.
B. memory address of pointer variable F.
C. memory address of alphabet F pointed by a pointer variable.
D. value pointed by the memory address stored in pointer variable F.

43. All of the following are important pillars of object-oriented programming (OOP) EXCEPT

A. abstraction.
B. typecasting.
C. encapsulation.
D. polymorphism.

44. The process of hiding irrelevant details from the user is called

A. abstraction.
B. inheritance.
C. encapsulation.
D. polymorphism.

45. Operator overloading and function overloading are the examples of

A. abstraction.
B. inheritance.
C. encapsulation.
D. polymorphism.

46. Which of the following statements is FALSE about constructors?

A. It has integer return type.
B. It initialises objects of a class.
C. It is a member function of a class.
D. It has a name which is same as the class name.

Use the given C++ program to answer Q.47, Q.48, Q.49 and Q.50.

```
#include<iostream>
using namespace std;

class Box {
  public:
    double length;
    double breadth;
    double height;
};

int main() {
  Box B1;
  Box B2;
  double volume = 0.0;

  B1.height = 7.3;
  B1.length = 8.5;
  B1.breadth = 10.0;

  B2.height = 12.0;
  B2.length = 13.8;
  B2.breadth = 15.0;

  volume = B1.height * B2.length * B1.breadth;
  cout << "Volume of Box1 : " << volume <<endl;

  volume = B2.height * B1.length * B2.breadth;
  cout << "Volume of Box2 : " << volume <<endl;
  return 0;
}
```

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47. The number of data members in **Box** class is

    A. 2
    B. 3
    C. 5
    D. 11

48. Names of objects in this program are

    A. B1 and B2.
    B. B1, B2 and volume.
    C. length, breadth and height.
    D. length, breadth, height and volume.

49. The number of data members that can be accessed inside as well as outside of the class is

	Access Inside the Class	Access Outside the Class
A	3	3
B	0	3
C	3	0
D	3	2

50. The output of the program is

Volume of Box1 : 1007.4 Volume of Box2 : 1530	Volume of Box1 : 620.5 Volume of Box2 : 2484
A	B
Volume of Box1 : 1007.4 Volume of Box2 : 2484	Volume of Box1 : 620.5 Volume of Box2 : 1530
C	D

ALTERNATE TO PRACTICAL (ATP: Questions 51-65)

Use the given program to answer Q.51, Q.52 and Q.53.

```
#include <iostream>
using namespace std;
int main( )
{
   int a, b=0;
   for(int c =1; c < 5; c++)
   {
      cin >> a;
      if(a > c)
      b = a + b;
   }
   cout << b + a;
   return 0;
}
```

51. Four sequences of values that are given as input to this program are shown in the options below.

    Which of the following options will give the output 8?

    A. 5, 4, 3 and 2
    B. 2, 3, 4 and 5
    C. 4, 3, 2 and 1
    D. 5, 3, 2 and 1

52. The following two statements of the program are changed.

```
for(int c =1; c < 5; c++)
if(a > c)
```

After the change in the program; when the following sequence is given as an input, then it outputs 74.

10, 11, 12, 13 and 14

The code lines that replace the original ones in the program to get this output are

for(int c =10; c < 15; c++) if(a < c)	for(int c =10; c <= 15; c++) if(a = = c)
A	B
for(int c =10; c < 15; c++) if(a >= c)	for(int c =10; c <= 15; c++) if(a <= c)
C	D

53. If the **for loop** code in this program is rewritten using the **do while loop**, then the CORRECT code is

int c = 1;
do
{
cin >> a;
if(a > c)
b = a + b;
c = c + 1;
}
while (c < 5);
	int c = 1;
	do
	{
	cin >> a;
	if(a > c)
	b = a + b;
	}
	while (c < 5);

A	B

int c = 1;
do
{
cin >> a;
if(a > c)
b = a + b;
c = c + 2;
}
while (c < 5);
	int c = 1;
	do while (c < 5);
	{
	cin >> a;
	if(a > c)
	b = a + b;
	c = c + 1;
	}

C	D

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54. When a program is executed, then the compiler shows error due to an invalid name of a variable used in the program. The variable name that caused the error is

A. answer5
B. answer 5
C. answer_5
D. _5answer

55. Consider the given program.

```
#include <iostream>
using namespace std;
int main()
{
   int y = 5;
   switch(y)
   {
   case 5: cout<<"Five";
   case 6: cout<<"Six";
           break;
   case 7: cout<<"Seven";
   default: cout<<"Default";
   }
   return 0;
}
```

The output of this program is

A. Five
B. Default
C. FiveSix
D. FiveSixSeven

56. An array named as **Age** is declared in a program and it has the following visual representation.

If each cell of the table represents the array's element, then the code to input value in the array's element represented by the shaded cell is

A. cin >> Age[5][3];
B. cin >> Age[5, 3];
C. cin >> Age[4][2];
D. cin >> Age[4, 2];

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Use the given program to answer Q.57 and Q.58.

Consider the given program.

```
#include <iostream>
#include <string.h>
using namespace std;
int main()
{
  char str1 [] = "Catch";
  char str2 [] = "Me";
  strcpy(str1,str2);
  cout << str1 << endl;
  cout << str2 << endl;
  return 0;
}
```

57. The output of this program is

Catch Me	Me Catch
A	B
Catch Catch	Me Me
C	D

58. The total number of elements in array **str1** after applying strcpy function is

A. 2
B. 3
C. 5
D. 6

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Use the given program to answer Q.59 and Q.60.

```
#include <iostream>
using namespace std;
int P(int);
int main( )
{
   int Q = P(24);
   cout << Q + 3;
   return 0;
}
int P(int Q)
{
   int P = 1;
   P = --Q % 5;
   return(P);
}
```

59. A statement from this program that shows the declaration of the user-defined function **P( )** is

    A. int P = 1;
    B. int P(int);
    C. int P(int Q)
    D. int Q = P(24);

60. The output of the given program is

    A. 3
    B. 6
    C. 26
    D. 27

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Use the given C++ program to answer Q.61, Q.62 and Q.63.

```
#include <iostream>
using namespace std;
class X1
{
   private:
      int num1 = 5, num2 = 6;
      float num3;
   public:
   int y;
     int fnc(int x)
     {
       num3 = num1 * num2  / x;
       return num3;
      }
};
 int main()
 {
     X1 p, q;
     float res;
     p.y = p.fnc(4);
     q.y = q.fnc(15);
     res = p.y + q.y;
     cout << "Answer is: " << res;
     return 0;
 }
```

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61. The statement(s) that show(s) the declaration of object(s) in this program is/ are

p.y = p.fnc(4); q.y = q.fnc(15);	X1 p, q;
A	B
int num1 = 5, num2 = 6; float num3;	int fnc(int x)
C	D

62. The number of data members that can be accessed outside of the class is

A. 1
B. 2
C. 3
D. 5

63. The output of this program is

Answer is: 9	Answer is: 9.5
A	B
Answer is: 10	Answer is: 10.5
C	D

Use the given program to answer Q.64 and Q.65.

```
#include <iostream>
using namespace std;
int main()
{
int x, *y;
x = 50;
y = &x;
cout << &y << " " << "0x6011c5"  << endl;
cout << "0x601176" << " " << y << endl;
return 0;
}
```

64. Following is the output of this program.

```
0x6011a0 0x6011c5
0x601176 0x601198
```

The memory address of the variable **x** is

A. 0x6011a0
B. 0x6011c5
C. 0x601176
D. 0x601198

65. To get the output 50, the statement that should replace both the **cout** statements in the program is

A. cout << *x;
B. cout << *y;
C. cout << &x;
D. cout << &y;

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END OF PAPER

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